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Level 4, 2nd Sem / DVOC (Medical Imaging Tech.)

Subject : Physics and Technology in Imaging

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

- Q.1 The majority of useful x-ray beam is composed of mainly
- a) Secondary radiation
 - b) k-rays
 - c) Brem's rays
 - d) Characteristics rays
- Q.2 A three phase, 12 pulse generator utilizes _____ pulses per second
- a) 6
 - b) 12
 - c) 360
 - d) 720
- Q.3 What is the primary purpose of glass housing
- a) Protect anode assembly
 - b) Provide vacuum
 - c) Protect cathode assembly
 - d) Kilo-voltage

Q.4 Acceleration of electron occurs when _____ is applied to the filament.

- a) Watt b) Joules
- c) milli-amperage d) None of the above

Q.5 The primary purpose of focusing cup is

- a) Narrow the electron cloud before acceleration towards target
- b) house the anode
- c) rotate the anode by Induction Motor
- d) transform electrons to x-ray photons

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 _____, _____ radiation is also called roentgen rays.

Q.7 X-ray region stands between ultraviolet and _____ region.

Q.8 Who discovered X-rays.

Q.9 Radiation Hazards.

Q.10 Dosimetry

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Write a note on various physical quantity and their units used in Radiology.

Q.12 Explain Radiation intensity-exposure.

Q.13 Explain relation between absorbed dose and equivalent dose.

Q.14 Write a note on scintillation detectors.

Q.15 Explain G.M counters with diagram.

Q.16 Write a note on fluroscopic screen accessories.

Q.17 Write a note on Factors affecting spatial resolution.

Q.18 Write a note on Properties of X-rays.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain Characteristic Radiation and Bremsstrahlung Radiation .

Q.20 Explain the types of x-ray interaction with matter.